

Disarmament and International Security Committee

Anti-Satellite Weapons and the Militarization of Outer Space

Backgrounder Guide

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Director's Letter

Dear Prospective Delegates,

My name is Marie Harmsworth, and it is my distinct honour to serve as your director for the Disarmament and International Security Committee at SPAMUN. Alongside my chair, Renie Zhu, and my assistant director, Abigail Rozmus, we are delighted to welcome you to Southpointe Academy Model United Nations' eighth iteration. Whether this is your very first conference or your last, I sincerely hope that you are ready to contribute your efforts, speeches, and funny speech starters to create a flourishing and fruitful debate environment.

As a senior at Lord Byng Secondary School in my fourth and final year of MUN, I have been working tirelessly to ensure an invigorating and exhilarating experience for delegates to fruitfully engage in. Model UN has served as a cornerstone of my high school experience. Though I began MUN as a nervous delegate who spoke a grand total of one time at my first conference, I rapidly developed a growing passion for this captivating sport. I strive to create an equally engrossing experience for each and every delegate who enters my committee room.

At this iteration of SPAMUN, DISEC will be exploring anti-satellite weapons and the militarisation of outer space. With the rapidly evolving, tumultuous state of our global political environment, this topic is quickly becoming critical to global peace.

As a committee, your task is simple: to debate, discuss, and collaborate to develop a comprehensive and pragmatic resolution paper which addresses all facets of anti-satellite weaponry. As a delegate, your role is to bring your ideas on how to multilaterally solve the dire concern of ASATs alongside your fellow delegates. Position papers are due on October 16th, 10 PM are highly encouraged and will be mandatory for award consideration.

I wish that all delegates not only read the backgrounder but also engage in their own research to grasp this topic to the fullest extent. I implore each and every one of you to enter the committee room with excitement, passion, and a readiness to collaborate with your peers. The only way that we as a committee will solve this crisis is if we enter SPAMUN with pragmatic, logical, and solution-oriented perspectives, which will ultimately benefit all countries involved.

If you have any questions or concerns, please contact us at disecspamun@southpointe.ca. I wish you all the best in your research and preparation. See you all in October!

Sincerely,

Marie Harmsworth

Director of the Disarmament and International Security Committee

Committee Description

Established in 1946, the Disarmament and International Security Committee serves as the first United Nations General Assembly. As one of the six primary committees, it was developed as an international platform where nations worldwide discuss issues regarding peace and security, emphasizing the principles of disarmament and international regulation of weaponry and arms. DISEC has stood behind some of the world's most instrumental actions surrounding many arms and security treaties. For instance, the Biological Weapons Convention (BWC) protects civilians worldwide from grave danger, violence, and turmoil.

Every single human on this earth has been touched by peace and security concerns. While many of us in the Lower Mainland have never truly experienced war firsthand, we must remain cognizant of the hundreds of millions of humans suffering from security threats worldwide. It is imperative that we, as a committee,e recognize the role we have as committee members to protect our societies from the pain and terror that so many people experience in war.

In the present day, the Disarmament and International Security Committee continues to stand as one of the principal UN bodies that aid peace and security worldwide. While resolutions created in DISEC continue to be non-binding to this day, the committee's ability to encourage and support nations in implementing pragmatic solutions can be effective if done properly. As delegates in this committee, it is important to ensure that your solutions are not only of sound logic but are created bearing in mind the optional nature of such resolutions and develop them in such a way that it is favourable for every nation to implement these aforementioned formal recommendations.

Topic Overview

With the 1957 launch of Sputnik 1, anti-satellite weapons and the weaponization of outer space are a largely contemporary issue that has rapidly become one of the forefront geopolitical spatial challenges. In the past, the exploration of outer space and the discovery of the vast universe beyond Earth were largely a scientific domain. But in recent decades, the domain of space has transformed into a fervent battle between numerous global hegemons. The weaponization of outer space—defined as the utilization of militaristic technology beyond Earth's atmosphere—can be boiled down to two key issues: anti-satellite weaponry and the imminent risk of Kessler Syndrome, and the outdated and blatantly inapplicable international frameworks in place to monitor the geopolitical and militaristic realm of outer space.

In this modern era, we as a society are deeply and unequivocally reliant on technology—satellites being one of the most imperative things keeping the world running as it does in this time. Within the past century or so, there has been a massive global shift away from traditional analog systems—exchanging them for digital networks, which rely heavily on satellites to keep these systems alive and running. Satellites have fundamentally transformed the way society operates. From the inception of Sputnik 1 to this day, satellite technology has served as a landmark technological development that has allowed human society to evolve rapidly into a highly connected, rapidly developing network. It has allowed substantive economic growth through modern technologies, the internet, and wifi, and an overall greater level of connectivity. Scientific research has also been revolutionized by the exploration of space and the use of satellite technologies, leading to major developments in scientific knowledge. Nowadays, satellites play an integral role in one's life—from calling a friend after work to turning on the TV, satellites appear in one's day in more ways than possibly imaginable. In essence, satellites have served as a catalyst for some of the human race’s most integral social, technological, and economic developments.

While it seems only logical that humans ought to capitalize on one of the world's greatest modern achievements, it is not as simple as it seems. As humans develop increasing dependence on these technologies, they can often become targets of attack. In this case, the destruction of powerful satellites is increasingly being threatened as a means of outer space militarization. With an entire nation's wifi networks, phone lines, or weather prediction systems hinging on these technologies, one brutal collision can bring havoc to a whole nation. Anti-Satellite weaponry, in short, is the weapons and systems that damage or destroy satellites.

Anti-satellite weaponry can be divided into two main categories: kinetic and non-kinetic. Kinetic weapons are those that destroy through means of physical disruption, collisions, or other tangible damage. On the other hand, non-kinetic weapons rely on cybersecurity threats—jamming signals, hacking, or disabling satellites to the point where they are unusable. The common

denominator: both kinetic and non-kinetic ASATs alike have risen to pose grave risks to modern society.

The Outer Space Treaty is a 1967 international legal framework that strives to protect the global society from the dangers of outer space militarization. This treaty strikes a balance between exploration and exploitation. On one hand, it protects outer space as a scientific domain that ought to be used to benefit nations in the interest of mankind. On the militarization side of it, however, it struggles to establish substantive or targeted regulations that safeguard nations from the potential damages of anti-satellite weaponry—whose ability to destroy, disrupt, and damage a nation's infrastructure has the potential to cause grave danger. In today's tumultuous geopolitical climate, there is a rapidly growing interest in strengthening ASAT technologies and a simultaneous urge to strengthen the inadequate legal regulations that currently surround them.

As a committee at SPAMUN 2026, the Disarmament and International Security committee must strive to bridge the gap between the archaic and frankly borderline irrelevant existing international frameworks and develop pragmatic and effective solutions. While not every nation in this battle holds the same stake in the space race, the collaboration and cooperation of delegates will be what protects the well-being and sacrosanct right of every nation to profit as a result of space technology. Space is not a warzone, and it is up to this committee to ensure it stays that way.

Timeline of Events

Cold War Space Era

October 4th, 1957—The USSR launches the world's first artificial satellite into Earth's orbit. Amid the chilling Cold War climate, its launch ushered in a slew of political and technological fears—marking the beginning of the space race.

July 29th, 1958—The American President Eisenhower oversaw the creation of a civilian space agency in the United States, signing the National Aeronautics and Space Act of 1958 on this day. On October 1st that same year, the National Aeronautics and Space Administration (NASA) officially opened.

October 13, 1959—The final test launch for the Bold Orion missile—an American air-launched anti-ballistic missile—was launched as an anti-satellite weapon. The first recorded incident of an ASAT vehicle coming within 4 miles of a satellite.

Expanded international cooperation and regulations in the late 1900's

January 27th, 1967—The Outer Space Treaty (OST), outlining the right to exploration of outer space and underscoring that space is free for exploration and cannot be claimed, as well as reaffirming that weapons of mass destruction could not be used in orbit, was signed and later put into effect the following October.

March 23rd, 1983—The Strategic Defence Initiative (SDI) was proposed by Ronald Reagan, the American president at the time, as a strategic system against nuclear attacks specifically from the Soviet intercontinental ballistic missiles.

Early 2000s–Modern developments

January 11th, 2007—The government of China conducted an ASAT weapon test—destroying a Chinese government satellite in space. This alarmed many with concerns about space debris and Kessler syndrome.

March 27th, 2019—Mission Shakti was an Indian test launch for an anti-satellite weapon whose target was a low Earth orbit vehicle. This event was known to raise great concerns regarding Kessler syndrome and debris.

November 15th, 2021—A Russian test of their "Nudol" ASAT system was launched into space, creating over 1500 fragments of debris in space, which posed a hazard for the International Space Station and other spacecraft alike for years.

April 12th, 2022—The United States instituted a unilateral ban on testing anti-satellite weapons. This was a landmark announcement showcasing an increased attention to sustainability in outer space.

Historical Analysis

On October 4th, 1957, a small object weighing only 83.6 kilograms and no larger than a beach ball was launched into space for the very first time. While it may seem insignificant now, this was the launch of a Soviet satellite—Sputnik 1—the world's first ever satellite. In the midst of the Cold War's frigid climate, this seemingly insignificant action launched a wave of terror and fright throughout the world. This vast demonstration of pure technopolitical supremacy triggered a paradigm shift in the geopolitical powers of the two global hegemons at the time, the USSR and the United States, an action that initiated the infamous space race.

At its core, Sputnik 1 was a scientific exploration whose success was considered a landmark technical achievement for space technology at the time. Though the demonstration of this vast scientific superiority raised a question for the Americans as well as other concerned nations: if they can launch Sputnik, what else can they do?

A second wave of terror hit upon the launch of a subsequent Sputnik 2 on November 3rd, 1957, transporting the first living creature—a stray dog from the streets of Moscow named Laika—into space for the first time.

These actions by the USSR sent the United States into a state of hysteria, revealing the "missile gap" and its shocking proficiency in the development of intercontinental ballistic missiles (ICBM)—the very technology used to launch Sputnik 1 and a technology with the potential to send nuclear warheads to the United States. As a response, the Eisenhower Administration in the States at the time quickly moved to develop the National Aeronautics and Space Administration (NASA) to parallel the USSR's spatial superiority and battle for technological superiority.

It is important to understand that the space race in fact had little to do with scientific innovation. Largely, it was about the militaristic and technological superiority of a nation's defence systems. The hysteria largely surrounded one simple question: If a nation has the technology to launch into space, what else can it launch? Specifically, civilians and governments alike were deeply concerned about intercontinental ballistic missiles, which many people feared would bombard the states and destroy nations as a whole.

As a result of these nationwide fears, the American government launched a bold Orion—an air-launched ballistic missile developed by the Martin Company in Maryland. While it failed to intercept the Explorer VI satellite, it managed to arrive within 4 miles—one, a remarkable achievement for the time.

It was only in 1967 when the Outer Space Treaty—a multinational agreement protecting space as an arena for technological research and solidifying limitations on the use of Weapons of Mass Destruction in Earth's orbit—was put into force. Decades later, this treaty continues to serve as the foundation for legal regulations regarding space, its exploration, and exploitation.

The Strategic Defence Initiative (SDI) was later proposed by the then American President Ronald Reagan as a defence against nuclear attacks—largely as a result of their fear of the Soviet Union's Intercontinental Ballistic Missiles (ICBMs).

For the next few decades, the two global hegemons at the time, the United States and Russia, continued developing anti-satellite technology and sharpened their scientific and militaristic blades. Other nations began to experiment in the anti-Satellite and space militarization waters as well. Hoping to sharpen their militaristic powers, the People's Republic of China conducted their first successful anti-satellite weaponry test by destroying one of its own satellites. China's 2007 anti-satellite weapon testing raised the international community's concern about Kessler Syndrome. However, the implications of this action were far more than minor, with the physical impact of satellite destruction fragmenting the satellite into over 1500 particles floating around Earth's orbit.

Kessler Syndrome exists as a chain reaction theory for particles in Earth's orbit. It is the paramount threat to long-term space usability. Essentially, if too many particles enter Earth's orbit, it could become so densely packed with fast-moving particles that launching any item, whether more ASATs, satellites, or crewed missions, would be impossible for decades. China's 2007 ASAT testing was a landmark case that brought concerns of Kessler Syndrome throughout the international community and serves as a testament to the dangers of anti-satellite weaponry and its testing.

Another nation that has delved into space militarization technology was the Mission Shakti, an Indian mission that showcased the nation's technological and scientific prowess. Launched over the Bay of Bengal, it opted for a direct ascent kinetic kill method—physically colliding with a satellite in low Earth orbit. This action has solidified India as a global space power, demonstrating its capabilities and capacity as a nation while being far more cautious than China, prioritizing technology that would quickly dissolve into harmless particles for satellites and future space missions.

On November 15th, 2021, Russia conducted yet another mission with a direct-ascent anti-satellite weapon, which intercepted a satellite and resulted in the creation of 1500 new particles in low Earth orbit. This brought grave danger to astronauts on the International Space Station (ISS), who were forced to take emergency procedures following the interception. While this did demonstrate Russia's capabilities with the Russian "Nudols" ICBM capability, it further highlighted how reckless actions in space pose major risks to the sustainability of space technology.

Months later, the United States announced a ban on anti-satellite missile tests. This was a calculated attempt to serve as a model for responsible use of outer space and to prevent further debris in space. While this action was instituted far too recently to truly observe the full effects of the American ban, nations around the globe, as well as the European Union, have pledged to follow in the American footsteps.

Current Situation

Currently, Kessler Syndrome remains one of the top concerns facing the future of outer space usage. As of July 28th, 2026, there are 29020 objects larger than 10 cm in space; approximately 18363 of said objects serve as active satellites, while the remaining thousands are space debris, much of which lies in Low Earth Orbit (LEO)—within 2000 km of the Earth's surface. A multitude of nations have followed in the United States' footsteps and have begun implementing space debris elimination strategies. Notably, the European Space Agency (ESA)—a European Union Agency—had adopted a Zero Debris approach targeting the elimination of debris production in Earth's orbits by 2030. This action has caused the ESA to implement stricter guidelines to substantively minimize space debris in any and all future space launches by ensuring the safe disposal of objects, limiting in-orbit collisions, and preventing the intentional release of space debris. Similarly, the Japanese Aerospace Exploration Agency (JAXA) has in recent years begun developing commercial space debris removal technology. Alongside underscoring the importance of space debris production, it is also seeking to provide solutions by tracking unused space debris and ultimately removing it from orbit.

While actions to prevent Kessler Syndrome seem promising, there is also a multilateral shift moving from kinetic ASAT weapons to non-kinetic ones so as to prevent any further space debris from appearing in space. Non-kinetic ASAT weapons destroy satellites without a physical collision or aggressive contact—eliminating the creation of thousands of satellite particles floating around in space orbits. Instead of traditional kinetic anti-satellite weaponry, these weapons utilize lasers, microwaves, and blind optical sensors as an attempt to disable or damage satellite systems. This contemporary space technology is becoming increasingly popular due to its efficacy, without extrapolating the growing risk of Kessler syndrome.

Now, more and more nations are rapidly developing anti-satellite space technology—not only for use but also as a deterrent for their satellite attacks. Therefore, in recent years there has been an increased prevalence and release of what is known as "bodyguard satellites" designed to protect a nation's most valuable satellites. This protection can operate in a multitude of ways, often equipped with sensors and high maneuverability in order to detect risks and protect the satellite from any attacking ASATs. While this technology is new, it is critical that these rapidly evolving

innovations are considered when implementing actionable solutions into the disarmament and international security committee. The increased prevalence of these is an active demonstration of the growing militarization of outer space concurrent with the global actions against Kessler Syndrome.

Large actors in the militarization of outer space remain largely the same as in the past decades, with the United States, Russia, and China acting as the spatial hegemons. Viewing space as a largely unconquered and warfighting domain, China has implemented an aggressive strategy with the goal of becoming a global space power. Having launched over 1000 satellites and built a powerful infrastructure, it is increasingly becoming a major power in more scientific and militaristic space domains. On the other hand, the United States and Russia (as the former Soviet Union) have long battled over the monopoly that is space militarization. Both these nations have played an integral role in the development of global space technology. With the United States' 2019 establishment of the US Space Force—a powerful 6th branch of the US military—they continue to produce cutting-edge space technology and missile defence systems to this day. Russia, the third power in this frenzied battle for spatial superiority, has long had a history of outer space militarization. While battling sanctions and socioeconomic struggles in recent years following the Russian invasion of Ukraine, it continues to thrive as a leading global space power and remains a great power in this modern space age.

Ultimately, a new frontier of space technology is on the rise, and it is up to the delegates of the disarmament and international security committee to deliver an actionable and pragmatic solution that capitalizes on these growing technologies while preventing the further militarization of outer space.

UN/International Involvement

The prevention of an arms race in outer space (PAROS)—the 1981 resolution 36/97/C—reaffirmed the basic principles of the aforementioned 1967 Outer Space Treaty (OST); it largely addressed the weaponization of outer space—specifically, it heavily recommended the banning of Anti-Satellite Weaponry (ASAT). In theory, this resolution could have played an instrumental role in preventing militarization in outer space; in practice, however, the non-binding nature of the PAROS resolution alongside the humongous military potential of weaponizing outer space created a legal gap between what the treaty outlined and the implementation of real agreed-upon international laws.

Though the resolution failed to institute multinational solutions that banned ASATs, it was pivotal in fostering multilateral negotiations, which have greatly informed the future of outer space policy. For instance, the PAROS negotiations incentivized states to develop comprehensive positions on the issue and strengthened the international community's understanding of the militarization of outer space, its risks, and reasonable limits. The resolutions' development further created a general understanding regarding the precise language of the prohibition of the weaponization of outer space. Though none of these advancements developed multilateral space legislation in itself, they have largely served as a blueprint for future multilateral negotiations and will continue to do so for years to come.

Possible Solutions

Safeguarding and monitoring the further development of anti-satellite weaponry

Anti-Satellite weaponry already exists. And it will not cease to do so. This solution calls upon the international community to create legislation to ensure the safe and peaceful development of anti-satellite weaponry going forward. Through measures such as conferences and increased communication and transparency about space efforts, this solution looks favourably upon the prevention of further space debris. As well as the peaceful and globally monitored use of weapons in space. Delegations favouring this solution should strive to create pragmatic solutions which implement realistic and attainable legislation that would prevent the further development of unsafe ASAT weapons causing debris and a fear for the future of outer space.

Increasingly multilateral transparency and trust building

The largest controlling issue in the militarisation of outer space is trust. From the space race to now, the core of space technology is mistrust. Leaving spatial hegemons such as the United States and Russia to battle it out in space. By developing international trust-building measures and increasing space technology transparency, it would be instrumental to evolving future space development. Multilateral transparency helps support the peaceful exploration of space while fostering global stability and understanding. This solution heavily relies on the implementation of trust-building measures multilaterally and establishing legal legislation clarifying a mandate about intentions and international norms.

Developing Space Situational Awareness (SSA) Systems

Space Situational Awareness Systems—which track orbital debris—play an integral role in the monitoring of space objects and are instrumental in the avoidance of Kessler syndrome. With over 21,600 satellites having been sent into Earth's orbit to this day, the importance of developing these systems is becoming increasingly important to the continuation of space utilization for governments and companies alike. These technologies would be able to limit debris, minimize the breakup of space debris, de-orbit spacecraft, and implement collision avoidance technology for spacecraft and satellites. While this technology may seem genial, there

are a multitude of drawbacks – specifically surrounding its accuracy and reliability. If this technology is to be used to address the issue, to monitor objects in space and prevent overcrowding, governments must focus on advancing the technology of SSAs in order to predict future orbits, identify potential threats, and model space debris in the event of a collision. It would also be important to generate a return system so these very tracking systems do not become a part of the issues themselves.

Bloc Positions

Western Bloc

The Western bloc, containing nations such as the United States, Canada, and the European Union, stands for the banning of anti-satellite weapon testing. The American government first introduced a unilateral ban on the testing of direct-ascent anti-satellite weapons on April 12th, 2022. This was a landmark decision from the United States, modelling a far more sustainable method of outer space management. Since then, many nations, and even the European Union, have committed to this very same self-imposed law, marking a shift towards responsible use of outer space. While nations in this bloc continue to develop anti-satellite technology and compete within the arms race, they advocate for space situational awareness systems and sustainable use agreements, which ensure that space is not left unusable for centuries.

Non-Aligned Movement

The Non-Aligned Movement strongly advocates for space exploration in the interest of humankind by peaceful means. Nations in this group strongly oppose an outer space arms race and the use of space as a militaristic domain and instead recognize the importance of peace and security for humankind. The Non-Aligned Movement advocates for compliance measures within preexisting arms agreements and legal regimes. As a group, they heavily advocate for greater international cooperation within the scientific domain of outer space. Delegations within this group should strive to incorporate greater safeguards and accountability systems within preexisting legal agreements while increasing global transparency within the domain of space exploration.

CRINK Group

Nations in this bloc, typically including China, Russia, and Iran, generally support the militarization of outer space. While they recognize concerns such as Kessler Syndrome, their prioritization of militaristic and technological advancement takes priority within their interest in space exploration. Nations such as Russia and China strive for dominance in the militaristic domain of space and continue to develop anti-satellite weaponry to destroy both civilian and

military satellites. Nations in this bloc should carefully balance sovereignty with international frameworks to maintain space as a military domain.

Discussion Questions

1. In what ways can DISEC create actionable solutions without the power to implement legislation? How can DISEC incentivize nations to incorporate greater regulation regarding ASATs?

2. How can the development of Kessler syndrome be mitigated while nations continue to develop and test ASAT weaponry?

3. How can the international community combat the monopolization of ASAT technologies by global powers while preventing their development as a whole?

4. How does the global shift towards non-kinetic anti-satellite weaponry change the way that international treaties regarding outer space militarisation should be treated?

5. Should prior legislation such as the Outer Space Treaty and PAROS resolutions be modified or completely rejuvenated to fit today's geopolitical climate?

6. What is a nation's responsibility in outer space? To what extent should nations value the continued use of outer space for future generations?

Additional Sources/Works Cited

Committee Description

"DISEC: Disarmament & International Security Committee." *IMUNA | NHSMUN | Model UN*, https://www.imuna.org/nhsmun/nyc/committees/disec-disarmament-international-security-committee/. Accessed 3 Aug. 2026.

United Nations. "UN General Assembly - First Committee - Disarmament and International Security." *Un.Org*, 2020, https://www.un.org/en/ga/first/.

Topic Overview

"Dawn of the Space Age - NASA." *NASA*, 4 Dec. 2025, https://www.nasa.gov/history/dawn-of-the-space-age/.

Duff, Robbie. "From Exploration to Escalation: The Challenges of Space Weaponization | Atlas Institute for International Affairs." *Atlas Institute for International Affairs*, 20 May 2025, https://atlasinstitute.org/from-exploration-to-escalation-the-challenges-of-space-weaponisation/.

NSE. "Modern Dependencies: The Impact of Complex Systems on Society and Civilization." *New Space Economy*, 11 Feb. 2025,

https://newspaceeconomy.ca/2025/02/11/modern-dependencies-the-impact-of-complex-systems-on-society-and-civilization/#google_vignette.

"Satellite Impact on Society: The Broad Social and Economic Effects of Satellite Technology on Everyday Life." *Diverse Daily*, 8 Aug. 2024, https://diversedaily.com/satellite-impact-on-society-the-broad-social-and-economic-effects-of-satellite-technology-on-everyday-life/.

Swope, Clayton. "The Future of Military Power Is Space Power - Aerospace Security." *Aerospace Security*, 10 Apr. 2025, https://aerospace.csis.org/the-future-of-military-power-is-space-power/.

United Nations Office for Outer Space Affairs. "The Outer Space Treaty." *Unoosa*, 1966, https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introouterspacetreaty.html.

Timeline of Events

141.70.6.91. "Mission Shakti." *Bharatpedia*, 22 Mar. 2022, https://en.bharatpedia.org/wiki/Mission_Shakti.

"China's Anti-Satellite Weapon Test: Congressional Research Service : Free Download, Borrow, and Streaming : Internet Archive." *Internet Archive*, 20 June 2012, https://archive.org/details/370838-chinas-anti-satellite-weapon-test.

"Dawn of the Space Age - NASA." *NASA*, 4 Dec. 2025, https://www.nasa.gov/history/dawn-of-the-space-age/.

Gregersen, Erik. *Strategic Defense Initiative | Description, History, & Facts*. *Encyclopædia Britannica*, 2019, https://www.britannica.com/topic/Strategic-Defense-Initiative.

"High Frontier." *Google Books*, 2026, https://books.google.ca/books?id=cMgdYypcPc8C&pg=PA65&redir_esc=y#v=onepage&q&f=false.

Maksym Steblivskyi. "Russia's 'Satellite Killer' Missile Program: Secret Documents Reveal New Details." *UNITED24 Media*, 22 July 2026, http://united24media.com/investigations/russias-satellite-killer-missile-program-secret-documents-reveal-new-details-20982.

Stiles, Curtis. "Two Years After the ASAT Test Ban: A Realistic Assessment - Missile Defense Advocacy Alliance." *Missile Defense Advocacy Alliance*, 10 May 2024, https://www.missiledefenseadvocacy.org/space-news/two-years-after-the-asat-test-ban-a-realistic-assessment/.

United Nations Office for Outer Space Affairs. "The Outer Space Treaty." *Unoosa*, 1966, https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introouterspacetreaty.html.

Uri, John. "65 Years Ago: The National Aeronautics and Space Act of 1958 Creates NASA - NASA." *NASA*, 26 July 2023, https://www.nasa.gov/history/65-years-ago-the-national-aeronautics-and-space-act-of-1958-creates-nasa/.

Historical Background

BOLD ORION FACT SHEET | Spaceline. https://www.spaceline.org/cape-canaveral-rocket-missile-program/bold-orion/. Accessed 3 Aug. 2026.

"China's Anti-Satellite Weapon Test : Congressional Research Service : Free Download, Borrow, and Streaming : Internet Archive." *Internet Archive*, 20 June 2012, https://archive.org/details/370838-chinas-anti-satellite-weapon-test/mode/1up.

"Dawn of the Space Age - NASA." *NASA*, 4 Dec. 2025, https://www.nasa.gov/history/dawn-of-the-space-age/.

Erwin, Sandra. "U.S. Declares Ban on Anti-Satellite Missile Tests, Calls for Other Nations to Join." *SpaceNews*, 19 Apr. 2022, https://spacenews.com/u-s-declares-ban-on-anti-satellite-missile-tests-calls-for-other-nations-to-join/.

Gregersen, Erik. *Strategic Defense Initiative | Description, History, & Facts*. *Encyclopædia Britannica*, 2019, https://www.britannica.com/topic/Strategic-Defense-Initiative.

"India's Mission Shakti: Retrospect and Prospect | Center for the Advanced Study of India (CASI)." *Upenn.Edu*, 29 July 2019, https://casi.sas.upenn.edu/iit/shounakset.

Lewis, Robert. "Laika | Background, Spaceflight, & Facts." *Encyclopedia Britannica*, 15 Nov. 2017, https://www.britannica.com/animal/Laika.

Logsdon, John M. *R-7 | Missile and Launch Vehicle | Britannica*. *Encyclopædia Britannica*, 2015, https://www.britannica.com/technology/R-7.

"Russia Conducts Direct-Ascent Anti-Satellite Test." *IISS*, https://www.iiss.org/online-analysis/online-analysis/2021/11/russia-conducts-direct-ascent-anti-satellite-test/. Accessed 3 Aug. 2026.

Shepley, Nick. "The Foundational Fears: Sputnik, the 'Missile Gap,' and the Crisis of American Techno-Confidence - Explaining History Podcast." *Explaining History Podcast*, 23 Nov. 2025,

https://explaininghistory.org/2025/11/23/the-foundational-fears-sputnik-the-missile-gap-and-the-crisis-of-american-techno-confidence/.

The Editors of Encyclopedia Britannica. *Sputnik | Satellites | Britannica*. *Encyclopædia Britannica*, 2025, https://www.britannica.com/technology/Sputnik.

United Nations Office for Outer Space Affairs. "The Outer Space Treaty." *Unoosa*, 1966, https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introouterspacetreaty.html.

"Untitled Document." *Www.Nasa.Gov*, https://www.nasa.gov/history/monograph10/nasabrth.html. Accessed 3 Aug. 2026.

"What Is Kessler Syndrome: The Space Debris Chain Reaction." *ScienceInsights*, 11 Mar. 2026, https://scienceinsights.org/what-is-kessler-syndrome-the-space-debris-chain-reaction/.

Current Situation

"Chapter 7: The Final Frontier: China's Ambitions to Dominate Space." *U.S.-China Economic and Security Review Commission*, https://www.uscc.gov/sites/default/files/2025-11/Chapter_7--The_Final_Frontier_Chinas_Ambitions_to_Dominate_Space.pdf. Accessed 3 Aug. 2026.

Gabrichidze, Tekla. "The Militarization of Space: China and Russia vs. The United States." *Rondeli Foundation*, 29 July 2024, https://gfsis.org/en/the-militarization-of-space-china-and-russia-vs-the-united-states/#.

Gregersen, Erik. *Space Debris | Facts, Removal, & Research*. *Encyclopædia Britannica*, 2019, https://www.britannica.com/technology/space-debris.

"JAXA | First Target Image From Orbit: The Image of Space Debris From Commercial Removal of Debris Demonstration Phase I Has Been Released." *JAXA | Japan Aerospace Exploration Agency*, 2024, https://global.jaxa.jp/press/2024/04/20240426-2_e.html.

"JAXA | Safety on Earth and in Space." *JAXA | Japan Aerospace Exploration Agency*, https://global.jaxa.jp/projects/debris/index.html. Accessed 3 Aug. 2026.

NSE. "Bodyguard Satellites and the Emerging Market for On-Orbit Protection." *New Space Economy*, 11 June 2026, https://newspaceeconomy.ca/2026/06/11/bodyguard-satellites-and-the-emerging-market-for-on-orbit-protection/.

"Space Debris Statistics." *Orbitalradar.Com*, Orbital Radar, 7 Mar. 2026, https://orbitalradar.com/space-debris-statistics.

The European Space Agency. "Mitigating Space Debris Generation." *Www.Esa.Int*, https://www.esa.int/Space_Safety/Space_Debris/Mitigating_space_debris_generation. Accessed 3 Aug. 2026.

"What Are Anti-Satellite Weapons (ASAT)? - SatNow." *Satnow.Com*, 19 Sept. 2025, https://www.satnow.com/community/what-are-anti-satellite-weapons-asat.

UN Involvement

"Publication | Prevention of an Arms Race in Outer Space: Multilateral Negotiations' Effects on International Law." *Gcsp.Ch*, 2020, https://www.gcsp.ch/publications/prevention-arms-race-outer-space-multilateral-negotiations-effects-international-law.

Reitmann, Louis. "The History of PAROS: An Overview of Multilateral Efforts for Space Security." *Vcdnp*, 2025, https://vcdnp.org/wp-content/uploads/2025/09/VCDNP_The-History-of-PAROS_web.pdf

Possible Solutions

"Secure World Foundation." *Swfound.Org*, 2023, https://www.swfound.org/publications-and-reports/swf-preprint-transparency-and-confidence-building-measures-for-outer-space-activities.

Space Foundation. "Space Situational Awareness." *Space Foundation*, 30 Jan. 2023, https://www.spacefoundation.org/space_brief/space-situational-awareness/.

"What Is Space Situational Awareness? | Ansys." *Ansys.Com*, 2025, https://www.ansys.com/simulation-topics/what-is-space-situational-awareness.

Bloc Positions

"Blocked." *Europeanspaceflight.Com*, 2025, https://europeanspaceflight.com/eu-member-states-commit-to-not-test-direct-ascent-asat-systems/.

Gabrichidze, Tekla. *The Militarization of Space: China and Russia vs. The United States*. Rondeli Foundation, 29 July 2024, https://gfsis.org/en/the-militarization-of-space-china-and-russia-vs-the-united-states/#.

Listner, Michael J, and Michael J Listner. "Two Years After the ASAT Test Ban: A Realistic Assessment — Global Security Review." *Global Security Review*, 9 May 2024, https://globalsecurityreview.com/two-years-after-the-asat-test-ban-a-realistic-assessment/.

New York. "CLUSTER 4 - FULL VERSION Statement by Delegation of the Republic of Indonesia on Behalf of the Non-Aligned Movement Thematic Debate: 'Conventional Weapons.'" *United Nations General Assembly. First Committee 78Th Session*,

https://docs-library.unoda.org/General_Assembly_First_Committee_-Seventy-Eighth_session_(2023)/Indonesia_NAM.pdf.